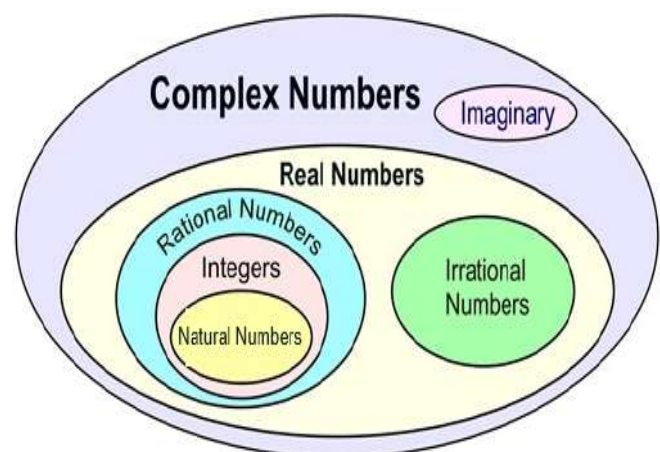


Complex analysis is a fundamental branch of mathematics that extends the concepts of calculus to the complex number system. This lecture will begin by exploring **Complex Numbers**, introducing their algebraic and geometric properties, including addition, multiplication, and conjugation. We will also discuss their representation on the complex plane, a crucial tool for visualizing and solving complex equations.

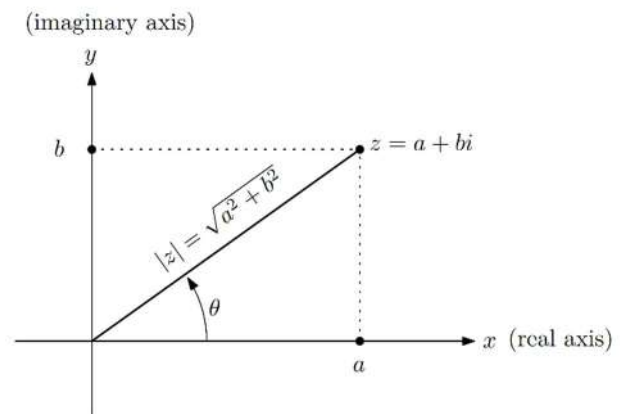
Next, we delve into the **Polar Form of Complex Numbers**, a powerful way of expressing complex numbers using magnitudes and angles. This representation simplifies operations like multiplication and division, setting the stage for an efficient approach to calculating **Powers and Roots** of complex numbers, using De Moivre's Theorem and other related techniques. Finally, we introduce **Complex Variables**, laying the foundation for deeper exploration of functions and integrals involving complex numbers, crucial in advanced applications like signal processing and fluid dynamics.

1. Complex Number.

A complex number is a number of the form $z = x + yi$ "**Cartesian Form**", where x and y are real numbers, and i is the imaginary unit, defined by $i^2 = -1$. In this expression, x is the **real part** of the complex number, and y is the **imaginary part**. Complex numbers are denoted as elements of the set $\mathbb{C}$, which extends the real number system $\mathbb{R}$ by including numbers that involve i .



Geometrically, complex numbers can be represented as points in the complex plane, where the **horizontal axis** represents the real part, and the **vertical axis** represents the imaginary part.



1.1 The Complex Conjugate.

The complex **conjugate** of a complex number $z = x + yi$, where x and y are real numbers, is denoted as $\bar{z}$ and is defined as $\bar{z} = x - yi$. Geometrically, the complex conjugate is the *reflection of z across the real axis* in the complex plane. This means that if z is represented as the point (x, y) , then $\bar{z}$ is the point $(x, -y)$. The real part of the complex number remains unchanged, while the imaginary part changes sign.

Properties of Complex Conjugate.

The complex conjugate has several useful properties that are fundamental in complex number operations:

1. **Product with Conjugate:** The product of a complex number and its conjugate is always a non-negative real number:

$$z\bar{z} = (x + yi)(x - yi) = x^2 + y^2 = |z|^2.$$

2. **Addition of Conjugates:** The sum of a complex number and its conjugate gives twice the real part of the number:

$$z + \bar{z} = (x + yi) + (x - yi) = 2x.$$

3. **Subtraction of Conjugates:** The difference between a complex number and its conjugate results in an imaginary number:

$$z - \bar{z} = (x + yi) - (x - yi) = 2yi.$$

4. **Conjugate of a Sum:** The conjugate of a sum of two complex numbers is the sum of their conjugates:

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

5. **Conjugate of a Product:** The conjugate of a product of two complex numbers is the product of their conjugates:

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$$

6. **Conjugate of a Quotient:** The conjugate of a quotient of two complex numbers is the quotient of their conjugates:

$$\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

7. **Multiplication Inverse z^{-1} :** given a complex number $z = x + yi \neq 0$, then its Multiplication Inverse is

$$z^{-1} = \frac{1}{z} = \frac{1}{x+yi} \frac{x-yi}{x-yi} = \frac{x}{x^2+y^2} - \frac{y}{x^2+y^2}i$$

These properties of the complex conjugate are instrumental in simplifying complex number expressions and solving complex equations.

1.2 The Modulus.

The **modulus** or **absolute value** of a complex number $z = x + yi$ is **the distance** from **the point** (x, y) in the complex plane to **the origin**. The modulus, denoted as $|z|$, is defined as

$$|z| = \sqrt{x^2 + y^2}.$$

This distance is always a non-negative real number. The modulus of a complex number plays a crucial role in understanding the magnitude of complex numbers, particularly in the polar form representation.

Properties of the Modulus

1. **Non-negativity:** The modulus of a complex number is always non-negative
 $|z| \geq 0$. The modulus equals zero if and only if $|z| = 0$.

2. **Multiplicative property:** The modulus of the product of two complex numbers is the product of their moduli:

$$|z_1 z_2| = |z_1| |z_2|$$

3. **Division property:** The modulus of the quotient of two complex numbers (when $z_2 \neq 0$) is the quotient of their moduli:

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

4. **Modulus of the conjugate:** The modulus of a complex number is equal to the modulus of its conjugate:

$$|z| = |\bar{z}|$$

5. **Triangle inequality:** The modulus satisfies the triangle inequality, which states that for any two complex numbers z_1 and z_2 :

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

These properties of the modulus are fundamental in complex analysis and have wide applications in various fields, including geometry, signal processing, and physics. The multiplicative and triangle inequalities, in particular, are critical in proving various theorems and solving complex problems involving complex numbers.

1.3 Complex Numbers Algebra.

The algebra of complex numbers is based on fundamental operations such as addition, subtraction, multiplication, and division. Given two complex numbers $z_1 = x_1 + y_1i$ and $z_2 = x_2 + y_2i$, where $x_1, y_1, x_2, y_2 \in \mathbb{R}$ and i is the imaginary unit with $i^2 = -1$, these operations follow specific rules.

1. **Addition:** The real and imaginary parts are added separately:

$$z_1 + z_2 = (x_1 + x_2) + (y_1 + y_2)i.$$

2. **Subtraction:** Similarly, involves subtracting the corresponding real and imaginary parts:

$$z_1 - z_2 = (x_1 - x_2) + (y_1 - y_2)i$$

3. **Multiplication:** To multiply two complex numbers, apply the distributive property:

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + (x_1 y_2 + x_2 y_1)i$$

This formula results from multiplying the terms and simplifying.

4. **Division:** involves multiplying both the numerator and denominator by the conjugate of the denominator to eliminate the imaginary part from the denominator.

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{x_1 + y_1i}{x_2 + y_2i} \\ &= \frac{(x_1 + y_1i)(x_2 - y_2i)}{(x_2 + y_2i)(x_2 - y_2i)} \\ &= \frac{(x_1 x_2 + y_1 y_2) + (y_1 x_2 - x_1 y_2)i}{x_2^2 + y_2^2} \\ &= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2}i\end{aligned}$$

This process ensures that the result is in the form $x + yi$, where x and y are real.

Example 1. Let $z_1 = 1 + 5i$ and $z_2 = -3 - 2i$. Evaluate the following.

1. $z_1 + z_2 = (1 + 5i) + (-3 - 2i) = -2 + 3i$
2. $z_1 - z_2 = (1 + 5i) - (-3 - 2i) = 4 + 7i$
3. $z_1 z_2 = (1 + 5i)(-3 - 2i) = 7 - 17i$
4. $\frac{z_1}{z_2} = \frac{1+5i}{-3-2i} = \frac{-13}{13} + \frac{-13}{13}i = -1 - i$
5. $\frac{1}{z_1} = \frac{1}{1+5i} = \frac{1}{26} - \frac{5}{26}i$

Example 2. Prove that.

1. $\frac{(2+i)^2}{3-4i} = 1$.
2. $z = 1 + i, z = -1 + i$ are roots of the equation $z^2 - 2zi - 2 = 0$.

Example 3. Do as required.

1. Find real and imaginary part of
 - a. $\frac{5}{4i-3}$
 - b. $\frac{3+4i}{1-i}$
 - c. $\frac{\sqrt{1+i}}{\sqrt{1-i}}$
2. Write $\frac{1}{2-3i} \frac{1}{1+i}$ in the Cartesian Form.
3. Verify $\frac{5}{(1-i)(2-i)(3-i)} = \frac{1}{2}i$.
4. Solve the equation $z^2 + z + 1 = 0$.
5. Find the multiplication inverse of $z = -2 + 3i$.

2. Polar Form of Complex Numbers.

The polar form of complex numbers provides a geometric perspective that is particularly useful for simplifying operations like multiplication, division, powers, and roots. Rather than expressing a complex number as $z = x + yi$, where x is the real part and y is the imaginary part, we represent the complex number in terms of its modulus r (magnitude) “*the distance from the origin to the point (x, y)* ”:

$$r = |z| = \sqrt{x^2 + y^2}$$

and angle θ (argument) “the angle made with the *positive real axis*”:

$$\text{either } \theta = \sin^{-1}\left(\frac{y}{r}\right), \theta = \cos^{-1}\left(\frac{x}{r}\right)$$

$$\text{or } \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

in polar form as:

$$z = r(\cos \theta + \sin \theta i)$$

or more compactly using Euler's formula:

$$z = re^{i\theta}$$

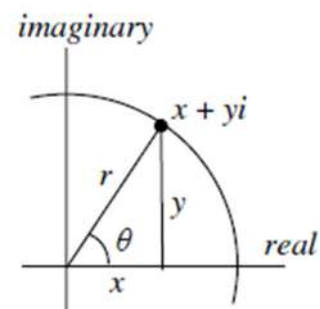
The polar form is particularly powerful when performing multiplication and division of complex numbers. If two complex numbers $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ are expressed in polar form, their product is:

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

Similarly, the division of two complex numbers is given by:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

This simplification of operations using polar form is one of its most significant advantages, as it reduces the complexity of calculations involving powers and roots of complex numbers.



Example 4. Do as required.

1. Express $\sqrt{3} + i$ in polar form.

$$\text{Modulus: } r = |z| = \sqrt{x^2 + y^2} = \sqrt{\sqrt{3}^2 + 1^2} = \sqrt{3 + 1} = \sqrt{4} = 2$$

$$\text{Argument: } \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\text{Polar form: } z = 2\left(\cos\frac{\pi}{6} + \sin\frac{\pi}{6}i\right) \text{ or Euler's formula } z = 2e^{\frac{\pi}{6}i}$$

2. Express $-3 + 3i$ in polar form.

$$\text{Modulus: } r = |z| = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + 3^2} = \sqrt{9 + 9} = \sqrt{18} = 3\sqrt{2}$$

$$\text{Argument: } \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{3}{-3}\right) = \frac{3\pi}{4}$$

$$\text{Polar form: } z = 3\sqrt{2}\left(\cos\frac{3\pi}{4} + \sin\frac{3\pi}{4}i\right) \text{ or Euler's formula } z = 3\sqrt{2}e^{\frac{3\pi}{4}i}$$

3. Express $z = 4\left(\cos\frac{\pi}{3} + \sin\frac{\pi}{3}i\right)$ in Cartesian form.

$$z = 4\frac{1}{2} + 4\frac{\sqrt{3}}{2}i = 2 + 2\sqrt{3}i$$

4. Multiply $z_1 = 2e^{\frac{\pi}{4}i}$ and $z_2 = 3e^{\frac{\pi}{6}i}$.

$$z_1 z_2 = 2e^{\frac{\pi}{4}i} 3e^{\frac{\pi}{6}i} = 6e^{\left(\frac{\pi}{4} + \frac{\pi}{6}\right)i} = 6e^{\frac{5\pi}{12}i}$$

5. Multiply $z_1 = 4e^{\frac{\pi}{3}i}$ and $z_2 = 2e^{\frac{\pi}{2}i}$.

$$z_1 z_2 = 4e^{\frac{\pi}{3}i} 2e^{\frac{\pi}{2}i} = 8e^{\left(\frac{\pi}{3} + \frac{\pi}{2}\right)i} = 8e^{\frac{5\pi}{6}i}$$

6. Divide $z_1 = 6e^{\frac{2\pi}{3}i}$ and $z_2 = 2e^{\frac{\pi}{3}i}$.

$$\frac{z_1}{z_2} = \frac{6e^{\frac{2\pi}{3}i}}{2e^{\frac{\pi}{3}i}} = 3e^{\left(\frac{2\pi}{3} - \frac{\pi}{3}\right)i} = 3e^{\frac{\pi}{3}i}$$

7. Divide $z_1 = 10e^{\frac{5\pi}{6}i}$ and $z_2 = 5e^{\frac{\pi}{4}i}$.

$$\frac{z_1}{z_2} = \frac{10e^{\frac{5\pi}{6}i}}{5e^{\frac{\pi}{4}i}} = 2e^{\left(\frac{5\pi}{6} - \frac{\pi}{4}\right)i} = 2e^{\frac{7\pi}{12}i}$$

3. De Moivre's Theorem (Powers and Roots).

De Moivre's Theorem is another important application of the polar form of complex numbers. It states that for any complex number $z = r(\cos \theta + \sin \theta i)$ and any integer n , the n^{th} power of z is $z^n = r^n[\cos(n\theta) + \sin(n\theta) i]$. This theorem is used extensively in calculating powers of complex numbers and finding roots.

Example 5. Find the powers to each of the following complex numbers.

1. $(1 + i)^5$

Modulus: $r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}$

Argument: $\theta = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$

Polar form: $z = \sqrt{2}\left(\cos \frac{\pi}{4} + \sin \frac{\pi}{4} i\right)$

or Euler's formula ?

$$z^5 = (\sqrt{2})^5 \left[\cos\left(5 \times \frac{\pi}{4}\right) + \sin\left(5 \times \frac{\pi}{4}\right) i \right] = 4\sqrt{2} \left(\cos \frac{5\pi}{4} + \sin \frac{5\pi}{4} i \right)$$

Cartesian form: $z^5 = -4\sqrt{2} - 4\sqrt{2}i$

2. $(2 + 2i)^3$

Modulus: $r = |z| = \sqrt{2^2 + 2^2} = 2\sqrt{2}$

Argument: $\theta = \tan^{-1}\left(\frac{2}{2}\right) = \frac{\pi}{4}$

Polar form: $z = 2\sqrt{2}\left(\cos \frac{\pi}{4} + \sin \frac{\pi}{4} i\right)$

or Euler's formula ?

$$z^3 = (2\sqrt{2})^3 \left[\cos\left(3 \times \frac{\pi}{4}\right) + \sin\left(3 \times \frac{\pi}{4}\right) i \right] = 16\sqrt{2} \left(\cos \frac{3\pi}{4} + \sin \frac{3\pi}{4} i \right)$$

Cartesian form: $z^3 = -16 + 16i$

3. $(3 - 4i)^4$

Modulus: $r = |z| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$

Argument: $\theta = \tan^{-1}\left(\frac{-4}{3}\right) = -\frac{\pi}{3} + 2\pi = \frac{5\pi}{3}$

Polar form: $z = 5\left(\cos\frac{5\pi}{3} + \sin\frac{5\pi}{3}i\right)$

or Euler's formula ?

$$\begin{aligned} z^4 &= 5^4 \left[\cos\left(4 \times \frac{5\pi}{3}\right) + \sin\left(4 \times \frac{5\pi}{3}\right)i \right] = 625 \left(\cos\frac{20\pi}{3} + \sin\frac{20\pi}{3}i \right) \\ &= 625 \left(\cos\frac{2\pi}{3} + \sin\frac{2\pi}{3}i \right) \end{aligned}$$

Cartesian form: $z^4 = 625 \times \frac{-1}{2} + 625 \times \frac{\sqrt{3}}{2}i = -312.5 + 312.5\sqrt{3}i$

Example 6. Find the roots to each of the following complex numbers.

1. The cube roots of $8 = 8^{\frac{1}{3}}$

Modulus: $r = |z| = \sqrt{8^2 + 0^2} = \sqrt{64} = 8$

Argument: $\theta = \tan^{-1}\left(\frac{0}{8}\right) = 0$

Polar form: $z = 8(\cos 0 + \sin 0 i)$

or Euler's formula ?

$$z^{\frac{1}{3}} = 8^{\frac{1}{3}} \left[\cos\left(\frac{1}{3} \times 0 + 2k\pi\right) + \sin\left(\frac{1}{3} \times 0 + 2k\pi\right)i \right]$$

where $k = 0, 1, 2$

then the 3 roots are:

$$2[\cos 0 + \sin 0 i] = 2,$$

$$2 \left[\cos\left(\frac{2\pi}{3}\right) + \sin\left(\frac{2\pi}{3}\right)i \right] = 2 \times \frac{-1}{2} + 2 \times \frac{\sqrt{3}}{2}i = -1 + \sqrt{3}i,$$

$$2 \left[\cos\left(\frac{4\pi}{3}\right) + \sin\left(\frac{4\pi}{3}\right)i \right] = 2 \times \frac{-1}{2} + 2 \times -\frac{\sqrt{3}}{2}i = -1 - \sqrt{3}i$$



2. The fourth roots of $16i = 16i^{\frac{1}{4}}$.
3. The square roots of -4 .

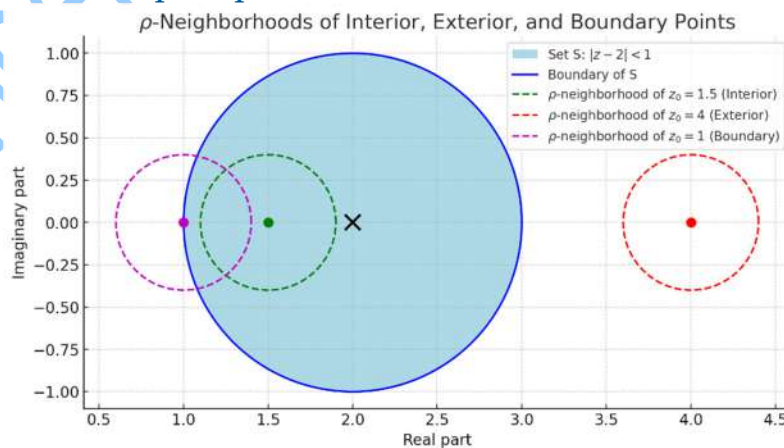
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2. Complex Plane Regions

A region in the complex plane is a set of points where various topological concepts are defined. The formal definitions of key topological terms are as follows:

1. **Neighborhood:** A neighborhood $N(z_0, \rho)$ of a point z_0 in the complex plane is a set of points that lie within a certain distance ρ from z_0 , i.e., $|z - z_0| < \rho$, where ρ is a positive real number. For example, the set $\{z: |z - 1| < 2\}$ is a neighborhood of $z = 1$.
2. Let S be a set of all complex numbers, and let z_0 be any complex number. Then:
 - a. z_0 is said to be an **interior point** of S if and only if there exists a ρ -neighborhood, denoted $N(z_0, \rho)$, that is completely contained within S .
 - b. z_0 is said to be an **exterior point** of S if and only if there exists a ρ -neighborhood, denoted $N(z_0, \rho)$, which contains no point of S , meaning $N(z_0, \rho) \cap S = \emptyset$.
 - c. z_0 is said to be a **boundary point** of S if and only if every ρ -neighborhood, denoted $N(z_0, \rho)$, contains points both inside S and outside of S .

Example7. Let S be the open disk defined by $S = \{z \in \mathbb{C}: |z - 2| < 1\}$, which is the set of all points in the complex plane inside the circle of radius 1 centered at $z = 2$.



Interior Point: Any point within the disk, such as $z_0 = 1.5$, is an interior point of S . This is because there exists a ρ -neighborhood around z_0 (e.g., a small disk with radius $\rho = 0.2$) that is fully contained within S .

Exterior Point: $z_0 = 4$ is an exterior point because there exists a ρ -neighborhood (e.g., with $\rho = 0.5$) around z_0 that contains no points of S , $N(z_0, \rho) \cap S = \emptyset$.

Boundary Point: $z_0 = 1$ is a boundary point because every ρ -neighborhood around z_0 will contain some points inside S (within the disk) and some points outside S (outside the disk).

3. **Open Set:** A set S is open if every point in S is an interior point. For example, the set $\{z: |z| < 1\}$ is an open set.
4. **Closed Set:** A set S is closed if it contains all its boundary points. The set $\{z: |z| \leq 1\}$ is closed because it includes the points on the boundary $|z| = 1$.
5. **Closure:** The closure $\bar{S}$ of a set S is the union of S and its boundary points. The closure of $\{z: |z| < 1\}$ is $\{z: |z| \leq 1\}$.
6. An open set S is **connected** if every pair of points within it can be joined by a polygonal path. This path consists of a finite number of line segments, joined end to end, and lies entirely within the set S . For instance, the disk $\{z: |z| < 1\}$ is connected.
7. **Domain:** A domain is an open and connected set. The set $\{z: |z| < 1\}$ is a domain.
8. **Region:** A region is a domain that may include none, some or all its boundary points. For example, both $\{z: |z| \leq 1\}$ and $\{z: |z| < 1\}$ are regions.

9. **Bounded/Unbounded:** A set is bounded if all points lie within a finite distance from the origin. The set $\{z: |z| < 1\}$ is bounded, while $\{z: |z| > 1\}$ is unbounded.
10. **Accumulation Point:** A point z_0 is an accumulation (or limit) point of a set S if every ρ -neighborhood $N(z_0, \rho)$ contains at least one point of S distinct from z_0 .

Example 8.

1. Let the set $S = \{\frac{1}{k}\}$, $k = 1, 2, 3, \dots$, thus, $S = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$.

- a. What are the interior points and the boundary points for the set S ?

There are no interior points, but all points in this set are boundary points.

(Since the set consists of discrete points, none of them have neighborhoods entirely contained within the set, so they are all boundary points.)

- b. Does the set S have a limit point?

Yes, $z_0 = 0$ is the unique limit point.

(As $k \rightarrow \infty$, $\frac{1}{k} \rightarrow 0$, so 0 is a limit point of the set.)

- c. Does the set S have a boundary?

Yes.

(The boundary of the set includes all the points in S , since no point has an open neighborhood that lies entirely within S .)

- d. Is the set S closed?

No, the set is not closed.

(A set is closed if it contains all its limit points. Since 0 is a limit point and is not included in S , the set is not closed.)

e. Is the set S open?

No.

(A set is open if every point in the set has a neighborhood entirely within the set. Since S consists of isolated points, it is not open.)

f. Is the set S connected?

No.

(The set S is a collection of discrete points, and therefore it is not connected.)

g. Is the set S a domain?

No.

(A domain is an open connected set, and S is neither open nor connected, so it is not a domain.)

h. What is the closure of the set S ?

The closure of S is $\bar{S} = \{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$.

(The closure of a set is the set itself plus its limit points. Since 0 is a limit point of S , the closure of S includes 0 along with all the points in S .)

2. Let the set $S = \{\frac{i}{k}\}$, $k = 1, 2, 3, \dots$ (answer a – h).

3. Complex Function.

A *complex variable* is a variable that takes values in the complex plane, typically denoted by $z = x + yi$, where x and y are real numbers, and i is the imaginary unit. Complex functions, defined on regions of the complex plane, play a central role in both theoretical and applied mathematics, with far-reaching implications in fields like physics and engineering.

A *complex function* f is a rule that assigns each $z \in D \subset \mathbb{C}$ a complex value $w = f(z)$. More formally, if $z = x + yi$, then $f(z) = u(x, y) + v(x, y)i$, where $u(x, y)$ and $v(x, y)$ are real-valued functions of the real variables x and y .

A function $w = f(z)$ is called *single-valued* in a region D if, for every $z \in D$, there is exactly one corresponding value of $f(z)$. Example: $w = z^2 + z + 1$, where for each value of z , the function provides only one value for w .

A function $w = f(z)$ is called *multi-valued* in a region D if, for some or all $z \in D$, there are multiple distinct values of $f(z)$ associated with the same z . Example: $w = z^{\frac{1}{3}}$, where for each z , the function can yield multiple distinct values, such as w_0, w_1, w_2 .

3.1 Complex Exponential Function

The complex exponential function is an extension of the real exponential function to complex numbers. For any complex number $z = x + yi$, where x and y are real, the complex exponential function is defined as:

$$e^z = e^{x+yi} = e^x [\cos(y) + \sin(y)i]$$

where e^x is the real exponential function, and $\cos(y) + \sin(y)i$ follows from Euler's formula $e^{yi} = \cos(y) + \sin(y)i$.

This formula shows that the complex exponential function combines both a real scaling factor e^x and a rotation in the complex plane by an angle y . The modulus of e^z is given by $|e^z| = e^x$. The argument of e^z is $\arg(e^z) = y$.

Example 9. find the Exponential of the following numbers.

1. $z = \pi i$

$$\begin{aligned} e^z &= e^{\pi i} \\ &= \cos(\pi) + \sin(\pi)i = -1 \end{aligned}$$

This is a famous result known as Euler's identity.

2. $z = 1 + \frac{\pi}{2}i$

$$\begin{aligned} e^z &= e^{1 + \frac{\pi}{2}i} \\ &= e^1 \left[\cos\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{2}\right)i \right] = ei \end{aligned}$$

3. $z = 2$

3.2 Complex Logarithmic Function

The complex logarithmic function is the inverse of the complex exponential function. For a non-zero complex number z , the complex logarithm, denoted by $\text{Log}(z)$, is defined as:

$$\text{Log}(z) = \ln|z| + \arg(z) i$$

where $\ln|z|$ is the natural logarithm of the modulus of z , and $\arg(z)$ is the argument (or phase) of z , which is multivalued, taking the form $\arg(z) = \theta + 2k\pi$ for any integer k .

Since the argument is multivalued, the complex logarithmic function is also multivalued, giving rise to different "branches." The principal value of the complex logarithm, denoted $\text{Log}(z)$, restricts the argument to $-\pi < \arg(z) \leq \pi$.

Example 10. find the logarithm of the following numbers.

1. $z = i$

$$\begin{aligned} \text{Log}(i) &= \ln|i| + \arg(i) i \\ &= \ln(1) + \frac{\pi}{2} i \\ &= \frac{\pi}{2} i \end{aligned}$$

2. $z = -1 + i$

$$\begin{aligned} \text{Log}(-1 + i) &= \ln|-1 + i| + \arg(-1 + i) i \\ &= \ln(\sqrt{2}) + \frac{3\pi}{4} i \\ &= \frac{\ln(2)}{2} + \frac{3\pi}{4} i \end{aligned}$$

3. $z = 4$

Example 11. Solve for z .

1. $e^z = 5$

$$\begin{aligned} z &= \log(5) = \ln|5| + \arg(5) i \\ &= \ln(5) + 0i = \ln(5) \end{aligned}$$

2. $e^z = -2$

$$\begin{aligned} z &= \log(-2) \\ &= \ln|-2| + \arg(-2) i \\ &= \ln(2) + \pi i \end{aligned}$$

the general solution being

$$\begin{aligned} z &= \ln(2) + (\pi + 2k\pi)i \\ &= \ln(2) + (1 + 2k)\pi i \end{aligned}$$

where k is any integer due to the periodicity of the logarithm.



3.3 Complex Trigonometric Function

Complex trigonometric functions are the natural extension of real trigonometric functions to the complex plane. Below are the formal definitions, along with their domains and ranges.

1. The complex **sine** function whose **domain** $z \in \mathbb{C}$ and **range** $\mathbb{C}$ is defined as:

$$\sin(z) = \frac{e^{zi} - e^{-zi}}{2i}$$

2. The complex **cosine** function whose **domain** $z \in \mathbb{C}$ and **range** $\mathbb{C}$ is defined as:

$$\cos(z) = \frac{e^{zi} + e^{-zi}}{2}$$

3. The complex **tangent** function whose **domain** $z \in \mathbb{C}, z \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\tan(z) = \frac{\sin(z)}{\cos(z)} = \frac{e^{zi} - e^{-zi}}{i(e^{zi} + e^{-zi})}$$

4. The complex **cosecant** function whose **domain** $z \in \mathbb{C}, z \neq k\pi, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\csc(z) = \frac{1}{\sin(z)} = \frac{2i}{e^{zi} - e^{-zi}}$$

5. The complex **secant** function whose **domain** $z \in \mathbb{C}, z \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\sec(z) = \frac{1}{\cos(z)} = \frac{2}{e^{zi} + e^{-zi}}$$

6. The complex **cotangent** function whose **domain** $z \in \mathbb{C}, z \neq k\pi, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\cot(z) = \frac{\cos(z)}{\sin(z)} = \frac{1}{\tan(z)} = \frac{i(e^{zi} + e^{-zi})}{e^{zi} - e^{-zi}}$$



Example 12. Simplify the following.

$$1. \sin(i) = \frac{e^{(i)i} - e^{-(i)i}}{2i} = \frac{e^{-1} - e^1}{2i}$$

$$= \frac{\frac{1}{e} - e}{2i} = \frac{1 - e^2}{2ei}$$

$$2. \cos(1+i) = \frac{e^{(1+i)i} + e^{-(1+i)i}}{2} = \frac{e^i e^{-1} + e^{-i} e^1}{2}$$

$$= \frac{e^{-1}[\cos(1) + \sin(1)i] + e^1[\cos(-1) + \sin(-1)i]}{2}$$

$$= \frac{e^{-1}[\cos(1) + \sin(1)i] + e[\cos(1) - \sin(1)i]}{2}$$

$$= \frac{(e^{-1} + e)\cos(1) + (e^{-1} - e)\sin(1)i}{2}$$

$$3. \tan(\pi i)$$

$$4. \sec(1+i)$$



3.4 Complex Inverse Trigonometric Function

Complex inverse trigonometric functions are the extension of the inverse trigonometric functions from the real numbers to the complex plane. below are the formal mathematical definitions of the most common complex inverse trigonometric functions, along with their domains and ranges.

1. The complex **inverse sine** function whose **domain** $z \in \mathbb{C}$ and **range** $-\frac{\pi}{2} \leq$

$\Re[\sin^{-1}(z)] \leq \frac{\pi}{2}$ is defined as:

$$\sin^{-1}(z) = -i \log \left(iz + \sqrt{1 - z^2} \right)$$

2. The complex **inverse cosine** function whose **domain** $z \in \mathbb{C}$ and **range** $0 \leq$

$\Re[\cos^{-1}(z)] \leq \pi$ is defined as:

$$\cos^{-1}(z) = \frac{\pi}{2} - \sin^{-1}(z) = -i \log \left(z + \sqrt{z^2 - 1} \right)$$

3. The complex **inverse tangent** function whose **domain** $z \in \mathbb{C}$ and **range** $-\frac{\pi}{2} <$

$\Re[\tan^{-1}(z)] < \frac{\pi}{2}$ is defined as:

$$\tan^{-1}(z) = \frac{i}{2} \log \left(\frac{1 - iz}{1 + iz} \right)$$

4. The complex **inverse cosecant** function whose **domain** $z \in \mathbb{C}, z \neq 0$ and **range**

$-\frac{\pi}{2} \leq \Re[\sin^{-1}(z)] \leq \frac{\pi}{2}$ is defined as:

$$\csc^{-1}(z) = \sin^{-1} \left(\frac{1}{z} \right) = -i \log \left(i \frac{1}{z} + \sqrt{1 - \frac{1}{z^2}} \right)$$

5. The complex **inverse secant** function whose **domain** $z \in \mathbb{C}, z \neq 0$ and **range** $0 \leq$

$\Re[\cos^{-1}(z)] \leq \pi$ is defined as:

$$\sec^{-1}(z) = \cos^{-1} \left(\frac{1}{z} \right) = -i \log \left(\frac{1}{z} + \sqrt{\frac{1}{z^2} - 1} \right)$$

6. The complex **inverse cotangent** function whose **domain** $z \in \mathbb{C}$ and **range** $0 < \Re[\cot^{-1}(z)] < \pi$ is defined as:

$$\cot^{-1}(z) = \frac{\pi}{2} - \tan^{-1}(z) = \frac{i}{2} \log \left(\frac{z-i}{z+i} \right)$$

Example 13. Simplify the following.

$$\begin{aligned} 1. \sin^{-1} \left(\frac{1}{2} \right) &= -i \log \left[i \left(\frac{1}{2} \right) + \sqrt{1 - \left(\frac{1}{2} \right)^2} \right] \\ &= -i \log \left[\frac{1}{2}i + \frac{\sqrt{3}}{2} \right] \end{aligned}$$

Now, let $z = \frac{\sqrt{3}}{2} + \frac{1}{2}i$, convert z into polar form $z = re^{i\theta}$:

$$r = \sqrt{\left(\frac{\sqrt{3}}{2} \right)^2 + \left(\frac{1}{2} \right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = 1$$

$$\theta = \tan^{-1} \left(\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6}$$

Thus,

$$z = e^{\frac{\pi}{6}i}$$

$$\begin{aligned} \sin^{-1} \left(\frac{1}{2} \right) &= -i \log \left[\frac{1}{2}i + \frac{\sqrt{3}}{2} \right] \\ &= -i \log \left[e^{\frac{\pi}{6}i} \right] \\ &= -i \left(\frac{\pi}{6}i \right) = \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} 2. \tan^{-1}(2+i) &= \frac{i}{2} \log \left(\frac{1-i(2+i)}{1+i(2+i)} \right) \\ &= \frac{i}{2} \log \left(\frac{2-2i}{2i} \right) = \frac{i}{2} \log \left[\frac{(2-2i)(-i)}{(2i)(-i)} \right] \\ &= \frac{i}{2} \log \left(\frac{-2i-2}{2} \right) = \frac{i}{2} \log(-1-i) \end{aligned}$$



$$|-1 - i| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{-1}{-1}\right) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$\tan^{-1}(2 + i) = \frac{i}{2}\left(\ln\sqrt{2} + \frac{5\pi}{4}i\right)$$

$$= \frac{i}{2}\left(\frac{1}{2}\ln(2) + \frac{5\pi}{4}i\right)$$

$$= \frac{1}{4}\ln(2)i - \frac{5\pi}{8}$$

3. $\cos^{-1}(2i)$

4. $\csc^{-1}[\csc(z)] = z$

3.5 Complex Hyperbolic Functions

Complex hyperbolic functions are the natural extensions of real hyperbolic functions to the complex plane. These functions are analogous to trigonometric functions but are based on hyperbolas instead of circles. Below are the formal mathematical definitions of the most common complex hyperbolic functions, including their domains, ranges, and relationships to trigonometric functions.

1. The complex **Hyperbolic sine** function whose **domain** $z \in \mathbb{C}$ and **range** $\mathbb{C}$ is defined as:

$$\sinh(z) = \frac{e^z - e^{-z}}{2}$$

2. The complex **Hyperbolic cosine** function whose **domain** $z \in \mathbb{C}$ and **range** $\mathbb{C}$ is defined as:

$$\cosh(z) = \frac{e^z + e^{-z}}{2}$$

3. The complex **Hyperbolic tangent** function whose **domain** $z \in \mathbb{C}, z \neq i\frac{\pi}{2} + k\pi, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\tanh(z) = \frac{\sinh(z)}{\cosh(z)} = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

4. The complex **Hyperbolic cosecant** function whose **domain** $z \in \mathbb{C}, z \neq k\pi i, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\operatorname{csch}(z) = \frac{1}{\sinh(z)} = \frac{2}{e^z - e^{-z}}$$

5. The complex **Hyperbolic secant** function whose **domain** $z \in \mathbb{C}, z \neq \frac{\pi}{2}i + k\pi, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\operatorname{sech}(z) = \frac{1}{\cosh(z)} = \frac{2}{e^z + e^{-z}}$$

6. The complex **Hyperbolic cotangent** function whose **domain** $z \in \mathbb{C}, z \neq k\pi i, k \in \mathbb{Z}$ and **range** $\mathbb{C}$ is defined as:

$$\coth(z) = \frac{\cosh(z)}{\sinh(z)} = \frac{1}{\tanh(z)} = \frac{e^z + e^{-z}}{e^z - e^{-z}}$$

There are key relationships between *hyperbolic* and *trigonometric* functions that hold in the complex plane:

- $\sinh(iz) = i \sin(z)$
- $\cosh(iz) = \cos(z)$
- $\tanh(iz) = i \tan(z)$
- $\operatorname{csch}(iz) = -i \csc(z)$
- $\operatorname{sech}(iz) = \sec(z)$
- $\coth(iz) = -i \cot(z)$

Example 14. Simplify the following.

1. $\cosh(\pi i) = \frac{e^{\pi i} + e^{-\pi i}}{2}$

Use Euler's formula:

$$e^{\pi i} = \cos(\pi) + i \sin(\pi) = -1 + 0 = -1$$

$$e^{-\pi i} = \cos(-\pi) + i \sin(-\pi) = -1 + 0 = -1$$

Hence,

$$\cosh(\pi i) = \frac{-1 - 1}{2} = \frac{-2}{2} = -1$$

$$\begin{aligned} 2. \sinh\left(\frac{\pi}{2} + \pi i\right) &= \frac{e^{\frac{\pi}{2} + \pi i} - e^{-\left(\frac{\pi}{2} + \pi i\right)}}{2} \\ &= \frac{e^{\frac{\pi}{2}} e^{\pi i} - e^{-\frac{\pi}{2}} e^{-\pi i}}{2} = \frac{-e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}}}{2} = ? \end{aligned}$$

3. $\tanh(1 + i)$

3.6 Complex Inverse Hyperbolic Functions

Complex inverse hyperbolic functions extend hyperbolic functions into the complex plane, linking exponential and logarithmic behavior. These functions provide key insights into complex analysis, with applications in areas like physics and engineering. Below are the formal definitions, including their domains, ranges, and relationships to trigonometric functions.

1. The complex **inverse Hyperbolic sine** function whose **domain** $z \in \mathbb{C}$ and **range** $\mathbb{C}$ is defined as:

$$\sinh^{-1}(z) = \log(z + \sqrt{z^2 + 1})$$

2. The complex **inverse Hyperbolic cosine** function whose **domain** $z \in \mathbb{C}, z \geq 1$ and **range** $\mathbb{C}$ is defined as:

$$\cosh^{-1}(z) = \log(z + \sqrt{z^2 - 1})$$

3. The complex **inverse Hyperbolic tangent** function whose **domain** $z \in \mathbb{C}, |z| < 1$ and **range** $\mathbb{C}$ is defined as:

$$\tanh^{-1}(z) = \frac{1}{2} \log\left(\frac{1+z}{1-z}\right)$$

4. The complex **inverse Hyperbolic cosecant** function whose **domain** $z \in \mathbb{C}, z \neq 0$ and **range** $\mathbb{C}$ is defined as:

$$\operatorname{csch}^{-1}(z) = \log\left(\frac{1}{z} + \sqrt{\frac{1}{z^2} + 1}\right)$$

5. The complex **inverse Hyperbolic secant** function whose **domain** $z \in \mathbb{C}, z \neq 0$ and **range** $\mathbb{C}$ is defined as:

$$\operatorname{sech}^{-1}(z) = \log\left(\frac{1}{z} + \sqrt{\frac{1}{z^2} - 1}\right)$$

6. The complex **inverse Hyperbolic cotangent** function whose domain $z \in \mathbb{C}, z > 1$ and range $\mathbb{C}$ is defined as:

$$\coth^{-1}(z) = \frac{1}{2} \log \left(\frac{z+1}{z-1} \right)$$

There are key relationships between complex inverse *hyperbolic* and *trigonometric* functions that hold in the complex plane:

- $\sinh^{-1}(iz) = i \sin^{-1}(z)$
- $\cosh^{-1}(iz) = i \cos^{-1}(z)$
- $\tanh^{-1}(iz) = i \tan^{-1}(z)$
- $\operatorname{csch}^{-1}(iz) = -i \csc^{-1}(z)$
- $\operatorname{sech}^{-1}(iz) = i \sec^{-1}(z)$
- $\coth^{-1}(iz) = -i \cot^{-1}(z)$

Example 15. evaluate the following.

$$\begin{aligned} 1. \sinh^{-1}(i) &= \log(i + \sqrt{i^2 + 1}) \\ &= \log(i + \sqrt{-1 + 1}) \\ &= \log(i) = \frac{\pi}{2}i \end{aligned}$$

$$2. \cosh^{-1}(2i)$$

Example 16. Verify the following.

$$1. \operatorname{Cosh}[\cosh^{-1}(2i)] = 2i$$

First evaluate $\cosh^{-1}(2i)$,

$$\begin{aligned} \cosh^{-1}(2i) &= \log \left((2i) + \sqrt{(2i)^2 - 1} \right) \\ &= \log(2i + \sqrt{-4 - 1}) \\ &= \log(2i + \sqrt{-5}) = \log(2i + \sqrt{5}i) \\ &= \log[(2 + \sqrt{5})i] = \log(2 + \sqrt{5}) + \log(i) \end{aligned}$$

$$= \log(2 + \sqrt{5}) + \frac{\pi}{2}i$$

Now, the entire RHS,

$$\begin{aligned} \text{Cosh}[\cosh^{-1}(2i)] &= \frac{e^{\cosh^{-1}(2i)} + e^{-\cosh^{-1}(2i)}}{2} \\ &= \frac{e^{\log(2+\sqrt{5}) + \frac{\pi}{2}i} + e^{-[\log(2+\sqrt{5}) + \frac{\pi}{2}i]}}{2} \end{aligned}$$

This can be simplified using Euler's formula:

$$\begin{aligned} \text{Cosh}[\cosh^{-1}(2i)] &= \frac{e^{\log(2+\sqrt{5})} [\cos(\frac{\pi}{2}) + \sin(\frac{\pi}{2})i] + e^{-\log(2+\sqrt{5})} [\cos(-\frac{\pi}{2}) + \sin(-\frac{\pi}{2})i]}{2} \\ &= \frac{(2 + \sqrt{5})i + \left(\frac{1}{2 + \sqrt{5}}\right)(-i)}{2} = \frac{(2 + \sqrt{5})i - \left(\frac{1}{2 + \sqrt{5}}\right)i}{2} \\ &= \frac{\left(2 + \sqrt{5} - \frac{1}{2 + \sqrt{5}}\right)i}{2} = \frac{\left[\frac{(2 + \sqrt{5})^2 - 1}{2 + \sqrt{5}}\right]i}{2} \\ &= \frac{\left(\frac{4 + 4\sqrt{5} + 5 - 1}{2 + \sqrt{5}}\right)i}{2} = \frac{\left(\frac{8 + 4\sqrt{5}}{2 + \sqrt{5}}\right)i}{2} \\ &= \frac{\left[\frac{4(2 + \sqrt{5})}{2 + \sqrt{5}}\right]i}{2} = \frac{4i}{2} \\ &= 2i \quad \text{LHS} \end{aligned}$$

2. $\sinh[\sinh^{-1}(1 + i)]$

4. Limit of a Complex Functions

Let $f(z)$ be a complex function defined in some neighborhood of a point z_0 , except possibly at z_0 itself. The complex function $f(z)$ is said to have a limit L as z approaches z_0 if for every $\epsilon > 0$, there exists a $\delta > 0$ such that whenever $0 < |z - z_0| < \delta$, it follows that $|f(z) - L| < \epsilon$. Symbolically, this is written as:

$$\lim_{z \rightarrow z_0} f(z) = L$$

This definition mirrors the concept of limits in real-valued functions, but the values z and z_0 are complex numbers, and distances are measured using the modulus of complex numbers.

To evaluate the limit of a complex function as $z \rightarrow \infty$, we use a substitution of the form $w = \frac{1}{z}$, which transforms the problem into evaluating the limit as $w \rightarrow 0$. This allows you to analyze the behavior of the function at infinity by considering the behavior at the origin. For example, if $\lim_{w \rightarrow 0} f\left(\frac{1}{w}\right) = L$, then:

$$\lim_{z \rightarrow \infty} f(z) = L$$

This method is particularly effective in handling rational functions or functions involving powers of z .

Example 17. Evaluate the following limits.

$$\begin{aligned} 1. \quad \lim_{z \rightarrow i} z^2 + 1 &= i^2 + 1 \\ &= -1 + 1 = 0. \end{aligned}$$

$$\begin{aligned} 2. \quad \lim_{z \rightarrow 2i} \frac{z^2 + 4}{z(z - 2i)} &= \lim_{z \rightarrow 2i} \frac{(z + 2i)(z - 2i)}{z(z - 2i)} \\ &= \lim_{z \rightarrow 2i} \frac{z + 2i}{z} = \frac{4i}{2i} = 2. \end{aligned}$$

$$3. \lim_{z \rightarrow \frac{\pi}{2}} e^{zi} = e^{\frac{\pi}{2}i}$$

$$= \cos \frac{\pi}{2} + \sin \frac{\pi}{2} i = i.$$

$$4. \lim_{z \rightarrow \infty} \frac{zi-2}{4z+i} = \lim_{z \rightarrow \infty} \frac{\frac{zi-2}{z}}{\frac{4z+i}{z}}$$

$$= \lim_{z \rightarrow \infty} \frac{i-\frac{2}{z}}{4+\frac{i}{z}} = \frac{i}{4}.$$

$$5. \lim_{z \rightarrow \infty} \frac{3z^2+2}{z^3+5z} = \lim_{z \rightarrow \infty} \frac{\frac{3z^2+2}{z^3}}{\frac{z^3+5z}{z^3}} = \lim_{z \rightarrow \infty} \frac{\frac{3}{z}+\frac{2}{z^3}}{1+\frac{5}{z^2}} = \frac{0}{1} = 0.$$

5. Continuity of a Complex Functions

A complex function $f(z)$ is said to be continuous at a point z_0 if small changes in z near z_0 result in small changes in $f(z)$ near $f(z_0)$. More precisely, $f(z)$ is continuous at z_0 if the following three conditions are satisfied:

1. The function must be defined at z_0 :

$$f(z_0) \text{ exists.}$$

2. The limit of the function must exist as z approaches z_0 :

$$\lim_{z \rightarrow z_0} f(z) \text{ exists.}$$

3. The limit of the function must equal the value of the function at z_0 :

$$\lim_{z \rightarrow z_0} f(z) = f(z_0).$$

If all three conditions are satisfied, then the function $f(z)$ is continuous at the point z_0 . A function is said to be continuous on a domain D if it is continuous at every point in D .

6. Derivative of a Complex Function.

In this section, we will define the derivative of a complex function, explore its use through examples, introduce the Cauchy-Riemann equations, discuss analytic functions, and conclude with a brief examination of Laplace's equations and harmonic functions.

The derivative of a complex function $f(z)$ at a point z_0 is defined by the following limit:

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Here, z is a complex variable, and Δz represents a small change in z . For the derivative to exist, the limit must be the same regardless of the direction from which Δz approaches zero. This condition is stricter than in real analysis and leads to important consequences, such as the Cauchy-Riemann equations, which we will discuss shortly.

Example 18. Using the definition of the derivative to find $f'(z)$ of the following.

1. $f(z) = z^2$

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{z^2 + 2z\Delta z + \Delta z^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{\Delta z(2z + \Delta z)}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} (2z + \Delta z) = 2z \end{aligned}$$

2. $f(z) = e^z$

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{e^{z+\Delta z} - e^z}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{e^z(e^{\Delta z} - 1)}{\Delta z} \\ &= e^z \lim_{\Delta z \rightarrow 0} \frac{(e^{\Delta z} - 1)}{\Delta z} = e^z \end{aligned}$$

3. $f(z) = \frac{1}{z}$

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{\frac{1}{z + \Delta z} - \frac{1}{z}}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{\frac{z - z - \Delta z}{z(z + \Delta z)}}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{-\Delta z}{z(z + \Delta z)\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{-1}{z(z + \Delta z)} = \frac{-1}{z^2} \end{aligned}$$

4.

The Cauchy-Riemann Equations (C – R Eq.)

Def. Let $f(z) = u(x, y) + iv(x, y)$, where $z = x + iy$. The equations

$$u_x(x, y) = v_y(x, y) \text{ — — — — — (1)}$$

$$u_y(x, y) = -v_x(x, y) \text{ — — — — — (2),}$$

are called the Cauchy – Riemann equations or Cauchy – Riemann conditions at $z = x + iy$.

Theorem (Cauchy – Riemann)

Suppose $f(z) = u(x, y) + iv(x, y)$, where $z_0 = x_0 + iy_0$. Then $\hat{f}(z_0)$ exists iff $u_x(x, y), u_y(x, y), v_x(x, y)$ and $v_y(x, y)$ exist and satisfy C-R Eq. 1 and 2, moreover $\hat{f}(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0) = v_y(x_0, y_0) - iu_y(x_0, y_0)$.

Ex. Let $f(z) = z^2 = x^2 - y^2 + 2ixy \rightarrow u = x^2 - y^2$, $v = 2xy$

$$u_x = 2x \text{ , } u_y = -2y \text{ , } v_x = 2y \text{ , } v_y = 2x$$

$\therefore u_x = v_y$ and $v_x = -u_y$,

$\therefore$ C – R Eq. are hold.

$\therefore \hat{f}(z) = u_x + iv_x = 2x + 2iy$ or $\hat{f}(z) = v_y - iu_y = 2x + 2iy$.

The Cauchy-Riemann Equations in Polar forms

Let $z = x + iy = re^{i\theta}$, $x = r\cos\theta$, $y = r\sin\theta$, and let $f(z) = u(x, y) + iv(x, y)$,

Using a chain rule and the def. of the derivative:

1- For r , $\hat{f}(z) \cdot \frac{\partial z}{\partial r} = u_r + iv_r$

since $z = re^{i\theta} \rightarrow \frac{\partial z}{\partial r} = e^{i\theta}$

$\therefore \hat{f}(z) \cdot e^{i\theta} = u_r + iv_r$ -----(1)

2- For θ , $\hat{f}(z) \cdot \frac{\partial z}{\partial \theta} = u_\theta + iv_\theta$, $\frac{\partial z}{\partial \theta} = rie^{i\theta}$

$\therefore \hat{f}(z) \cdot rie^{i\theta} = u_\theta + iv_\theta$ -----(2)

$\hat{f}(z) \cdot ire^{i\theta} = iru_r - rv_r$ -----(1) (x ir)

i.e. from 1' & 2 $\rightarrow iru_r - rv_r = u_\theta + iv_\theta$

$$u_{\theta} = -rv_r$$

$$v_{\theta} = ru_r \quad \text{C-R Eq. in polar form.}$$

Now what about the derivative $\hat{f}(z)$ in polar form ?

$$\hat{f}(z) = e^{-i\theta} [U_r(r, \theta) + iV_r(r, \theta)] \text{ ----- (3)}$$

$$= \frac{1}{r} e^{-i\theta} [V_{\theta}(r, \theta) - iU_{\theta}(r, \theta)] \text{ -----(4).}$$

Ex. Consider the function $g(z) = \sqrt{r} \cdot e^{i\frac{\theta}{2}}$, ($r > 0$, $0 < \theta \leq \pi$). show that $g(z)$ Has a derivative at each point in its domain of definition and $\dot{g}(z) = \frac{1}{2g(z)}$?

Sol. $g(z) = r^{\frac{1}{2}} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right) = r^{\frac{1}{2}} \cos \frac{\theta}{2} + i r^{\frac{1}{2}} \sin \frac{\theta}{2}$.

$\therefore u = r^{\frac{1}{2}} \cos \frac{\theta}{2}$, $v = r^{\frac{1}{2}} \sin \frac{\theta}{2}$

$$U_r = \frac{1}{2} r^{-\frac{1}{2}} \cos \frac{\theta}{2}, \quad U_\theta = -\frac{1}{2} r^{\frac{1}{2}} \sin \frac{\theta}{2}$$

$$V_r = \frac{1}{2} r^{-\frac{1}{2}} \sin \frac{\theta}{2}, \quad V_\theta = \frac{1}{2} r^{\frac{1}{2}} \cos \frac{\theta}{2}$$

$\therefore C - R$ Eq. are holds, $\therefore \dot{g}(z)$ exist.

$$\begin{aligned} \dot{g}(z) &= e^{-i\theta} [U_r + iV_r] = e^{-i\theta} \left[\frac{1}{2} r^{-\frac{1}{2}} \cos \frac{\theta}{2} + i \frac{1}{2} r^{-\frac{1}{2}} \sin \frac{\theta}{2} \right] \\ &= e^{-i\theta} \frac{1}{2\sqrt{r}} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right) = e^{-i\theta} \frac{1}{2\sqrt{r}} e^{i\frac{\theta}{2}} = \frac{1}{2\sqrt{r}} e^{-i\frac{\theta}{2}} = \frac{1}{2\sqrt{r} e^{-i\frac{\theta}{2}}} \\ &= \frac{1}{2g(z)}. \end{aligned}$$

Analytic Function

Def. A function f of the complex variable z is **analytic** at a point z_0 if its derivative exists not only at z_0 but at each point of a neighborhood of z_0 .

Def. A function f is said to be analytic in the D if it is analytic at each point of the domain D .

Ex. $f(z) = |z|^2$ is not analytic at any point except at $z = 0$

$f(z) = 3z - \frac{1}{3}z + 1$ is analytic everywhere.

Def. A function f is **entire** if it is analytic at each point of the complex plane.

Ex. $f(z) = e^z$ is entire,

$f(z) = \sum_{k=0}^n a_k z^k$ is entire.

Def. If a function f is not analytic at z_0 but it is analytic at each point of neighborhood of z_0 , then z_0 is called a singular point or a singularity of f .

Ex. $f(z) = \frac{\sin z}{z-2}$, the singular point is $z = 2$.

Harmonic Functions

Def. A real – valued function h of two real variables x and y is said to be harmonic in a given domain of the xy – plane if through that domain it has contains first and second partial derivatives and satisfies the partial differential equation :

$$h_{xx}(x, y) + h_{yy}(x, y) = 0 ,$$

this equation is called the Laplace's equation .

Def. If two given functions $U(x, y)$ and $V(x, y)$ are harmonic in a domain D and their first partial derivatives satisfy the $C - R$ Eq. Through D .

we say that V is the harmonic conjugate of U .

Theorem. A function $f(z) = U(x, y) + iV(x, y)$ is analytic in a domain D iff V is the harmonic conjugate of U .

Ex. If $u(x, y) = y^3 - 3x^2y$ is the *Re.* part of the analytic function $f(z)$, find the *Im.* Part (harmonic conjugate)?

Sol. $U_x = -6xy = V_y$, $U_y = 3y^2 - 3x^2$,

$$V_y = -6xy \rightarrow V = \int -6xy dy \rightarrow V = -3xy^2 + F(x)$$

$$V_x = -3y^2 + F'(x) = -U_y \rightarrow -3y^2 + F'(x) = -3y^2 + 3x^2 \rightarrow F'(x) = 3x^2$$

$$\therefore F(x) = x^3 + c.$$

$$\therefore V = -3xy^2 + x^3 + c.$$

Ex. Prove that $f(x, y) = e^x \cos y$ is a harmonic function ?

Sol. $\frac{\partial f}{\partial x} = e^x \cos y \rightarrow \frac{\partial^2 f}{\partial x^2} = e^x \cos y$

$$\frac{\partial f}{\partial y} = -e^x \sin y \rightarrow \frac{\partial^2 f}{\partial y^2} = -e^x \cos y$$

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = e^x \cos y + (-e^x \cos y) = 0.$$

$\therefore f$ is harmonic .

Ex. If $U(x, y) = e^x \cos y$, find the harmonic conjugate $V(x, y)$?

Sol. $U_x = e^x \cos y = V_y$, $U_y = -e^x \sin y$

$$V_y = e^x \cos y \rightarrow V = e^x \sin y + F(x)$$

$$V_x = e^x \sin y + F'(x) = -U_y \rightarrow e^x \sin y + F'(x) = -(-e^x \sin y)$$

$$F'(x) = 0 \rightarrow F(x) = c. \quad , \quad \therefore V = e^x \sin y + c.$$

Laplace's Equation in Polar form

$$U_{xx} + U_{yy} = 0 \approx r^2 U_{rr} + r U_r + U_{\theta\theta} = 0 \quad \text{-----(1)}$$

$$V_{xx} + V_{yy} = 0 \approx r^2 V_{rr} + r V_r + V_{\theta\theta} = 0 \quad \text{-----(2) Laplace's Eq. in polar form.}$$

Ex. Let $g(z) = \sqrt{r} e^{i\frac{\theta}{2}} \rightarrow U = \sqrt{r} \cos \frac{\theta}{2}, \quad V = \sqrt{r} \sin \frac{\theta}{2}$

$$U_r = \frac{1}{2} r^{-\frac{1}{2}} \cos \frac{\theta}{2} \rightarrow U_{rr} = \frac{-1}{4} r^{-\frac{3}{2}} \cos \frac{\theta}{2}$$

$$U_\theta = \frac{-1}{2} r^{\frac{1}{2}} \sin \frac{\theta}{2} \rightarrow U_{\theta\theta} = \frac{-1}{4} r^{\frac{1}{2}} \cos \frac{\theta}{2}$$

$$\therefore r^2 \left(\frac{-1}{4} r^{-\frac{3}{2}} \cos \frac{\theta}{2} \right) + r \left(\frac{1}{2} r^{-\frac{1}{2}} \cos \frac{\theta}{2} \right) - \frac{-1}{4} r^{\frac{1}{2}} \cos \frac{\theta}{2} = 0$$

$\therefore$ laplace's Eq. is hold. $\therefore U$ is harmonic .

Q. Let $g(z) = r^2 e^{i2\theta}$, prove that V is harmonic ? (H.W).